DEPARTMENT OF THE NAVY
NAVAL SURFACE WARFARE CENTER
INDIAN HEAD DIVISION
3767 STRAUSS AVENUE
SUITE 201
INDIAN HEAD, MD 20640-5150

J&A Number: FY25-FEB19-45
Code: 026

JUSTIFICATION AND APPROVAL
FOR USE OF OTHER THAN FULL AND OPEN COMPETITION

1. Contracting Activity

The Naval Sea Systems Command, Naval Surface Warfare Center Indian Head Division (NSWC IHD), Contracts Office Code 026.

2. Description of the Action Being Approved

Award of a sole source contract for a 600 MHz General Nuclear Magnetic Resonance (NMR) system, a 600 MHz Nitrogen NMR system, and additional accessories/software to:

Bruker Biospin Corporation
15 Fortune Drive
Billerica, MA 01821
CAGE Code: 54437

3. Description of Supplies/Services

NSWC IHD Research and Development (R) Department (R11) specializes in the synthesis and characterization of novel energetic materials, and as such requires the purchase of one (1) 600 MHz General NMR system and one (1) 600 MHz Nitrogen NMR system. The 600 MHz General NMR system purchase contains a Bruker iProbe Broad Band Fluorine Observation (BBFO) – 5mm, Avance Neo Console, Ascend Evo 600 MHz magnet, VT Gas Membrane Air Dryer, 5mm Spinners, BCU-II VT Gas Cooler, N_2 VT Gas Separator, 10KVA UPS, LN_2 Evaporator for low VT, software accessories, and general maintenance accessories. The 600 MHz Nitrogen NMR system contains a Bruker Prodigy BBO-H&F 5mm CryoProbe, Avance Neo Console, Ascend Evo 600 MHz magnet, VT Gas Membrane Dryer, 5mm Spinners, 10 L Buffer for BCU-II VT Gas Air Dryer, N_2 VT Gas Separator, BCU-I VT Gas Cooler, 10KVA UPS, 10mm Broad Band Observed (BBO) Probe with required adaptors, software accessories, and general maintenance accessories.

Upon receipt, the contractor will assemble the electronic components of the instruments and verify specifications via data collection of a known standard. Training of six (6) users is included as well as phone support for maintenance and troubleshooting.

The total net cost of this proposed contract action is estimated to be [redacted].

[redacted]

The Government minimum needs have been verified by the certifying technical and requirements personnel.

4. Statutory Authority Permitting Other Than Full and Open Competition.

This acquisition is being conducted under the authority of 10 U.S.C. 3204 (a) (1), Only One Responsible Source and No Other Supplies or Services will Satisfy Agency Requirements.

5. Rationale Justifying Use of Cited Statutory Authority.

Two Bruker 600 MHz NMR systems are required to characterize novel energetic materials. Both the 600 MHz General NMR system and the 600 MHz Nitrogen NMR system provide characterization data incapable of being collected by any other characterization method available on the market. New technologies developed by Bruker regarding their Nitrogen NMR system will enable the analysis of more sensitive energetic materials in a manner not previously available to NSWC IHD R11 Research Scientists. The auto-calibrate software accessories that Bruker provides with their NMR systems will also significantly decrease the labor hours needed to troubleshoot any maintenance issues.

Purchase of the Bruker 600 MHz General NMR system and the 600 MHz Nitrogen NMR system will be a significant upgrade from the current NMR, a Varian 300 MHz. The current equipment, Varian, that NSWC uses to complete the mission is no longer available as of 2014. To this end, Bruker has been at the forefront of NMR technologies for 64 years. They are the only manufacturer continually updating and improving their proprietary technology. Bruker is the sole manufacturer that has a very high intermediate frequency of 1.852 GHz for NMR signal generation and detection, which avoids any compromise with the local oscillator windows in regards to noise and decoupling leakage. They are the only manufacturer that has a console with 2 independent transceiver channels and support for up to kilowatt amplifiers. Their consoles have trademarked NMR Thermometer technology, which is the only real-time variable-temperature control that uses the actual internal sample temperature to regulate and control temperature in real-time during experiments.

They are the sole manufacturer that has a 600 MHz Magnet with a very low Helium evaporation rate (13mL He/hr) and 365-day helium hold time under stabilized conditions. Their proprietary probes have a wide temperature range allowing experiments to be run anywhere from -150C to +150C, which would allow scientists to run all possible samples that would be created. Their systems have standardized platform software (TopSpin4, which is utilized across all Bruker FT-NMR instrumentation.) This platform allows for easy transition between low field (down to 80 MHZ) and high field FT-NMR (up to 1.2 GHz).

The above equipment abilities will allow NSWC IHD R11 to have better data compiled that will improve the research and development of our energetic material, improving efficiencies and time to results. In addition, users in R11 are familiar with Bruker systems which avoid time, labor, and cost required for training.

Utilizing Bruker instrumentation allows us to access and retain records from decades of work. The labs utilize current Bruker instrumentation and software allowing for cross talk between instruments and allows resource sharing of IT equipment. The inability to use decades of research and development will cause additional time and money.

6. Description of Efforts Made to Solicit Offers from as Many Offerors as Practicable.

Market research was conducted, and no other manufacturer was found to meet the characterization needs nor the familiarity requirements with the operating software accessories for the NSWC IHD R11 Research Scientists.

A sources sought notice was posted to the System for Award Management website (SAM.gov) to satisfy requirements for Defense Federal Acquisition Regulation Supplement PGI 206.302-1(d). The sources sought notice was posted on 12 November 2024 and closed on 27 November 2024. Three (3) responses were received of which only one (1) provided a capability statement that was acceptable, and that was Bruker. The other submissions lacked the ability to produce the updated technology to produce the characterization data needed.

As required by FAR 5.203, a pre-solicitation notice was publicized on the SAM.gov on 04 December 2024. No responses were received. No additional market research was conducted because it was not practical, for the reasons discussed in paragraph 5 above.

7. Determination of Fair and Reasonable Cost.

The Contracting Officer has determined the anticipated cost to the Government of the supplies/services covered by this J&A will be fair and reasonable.

8. Actions to Remove Barriers to Future Competition.

For the reasons set forth in paragraph 5, NSWC IHD Contracts Department has no plans at this time to compete future contracts for the types of supplies/services covered by this document. If another potential source emerges, NSWC IHD Contracts Department will assess whether competition for future requirements is feasible.

CERTIFICATIONS AND APPROVAL

TECHNICAL/REQUIREMENTS CERTIFICATION

I certify that the facts and representations under my cognizance which are included in this Justification and its supporting acquisition planning documents, except as noted herein, are complete and accurate to the best of my knowledge and belief.

Technical/Requirements Cognizance:

[redacted]

LEGAL SUFFICIENCY REVIEW

I have determined this Justification is legally sufficient.

[redacted]

CONTRACTING OFFICER CERTIFICATION

I certify that this Justification is accurate and complete to the best of my knowledge and belief.

[redacted]

APPROVAL

Upon the basis of the above justification, I hereby approve, as Deputy Chief of the Contracting Office, the solicitation of the proposed procurement(s) described herein using other than full and open competition, pursuant to the authority of 10 U.S.C. 3204(a)(1).

LUTHER.RACHEL.NE [redacted]
WMAN [redacted]

Rachel N. Luther [redacted]
Deputy Chief of the Contracting Office